# Digital Assignment 1 (DA1) — Project Proposal & Data Preparation

## Project Title

**Cryptocurrency Transaction Fraud Detection: A Multi-Algorithmic Approach to Anti-Money Laundering on the Bitcoin Network**

## 1. Problem Statement and Objectives

Cryptocurrency's pseudonymous nature makes it an attractive channel for money laundering, ransomware payments, and dark-web market transactions. This project builds a classification pipeline to distinguish **licit** from **illicit** Bitcoin transactions using graph-derived transaction features, with the goal of supporting anti-money-laundering (AML) compliance workflows.

Two characteristics of this problem make it non-trivial and worth studying:

1. **Severe class imbalance** — illicit transactions make up only ~2% of labeled data, so naive accuracy-based evaluation is misleading; a majority-class classifier would already exceed 97% accuracy while catching zero fraud.
2. **Temporal structure** — transactions are timestamped across 49 sequential time steps, so a model must be evaluated on a time-aware split (train on earlier steps, test on later ones) to avoid data leakage and to reflect a realistic deployment setting where the model must generalize to future transactions it hasn't seen.

**Objectives:**

- Build a clean, preprocessed dataset from the raw Elliptic transaction graph
- Address class imbalance through SMOTE-based oversampling on the training set
- Compare multiple ML/DL algorithms (DA2) to identify which best detects illicit activity, prioritizing F1-score and Precision-Recall AUC over raw accuracy
- Deliver an interactive dashboard (DA3) that surfaces model predictions and explains *why* a transaction was flagged

## 2. Dataset Identification and Justification

**Dataset:** The Elliptic Data Set (Elliptic Co., released via Kaggle: kaggle.com/datasets/ellipticco/elliptic-data-set)

**Why this dataset:**

- It is a **real** Bitcoin transaction graph — not synthetic — released by Elliptic, a blockchain analytics company, in collaboration with researchers from the MIT-IBM Watson AI Lab

- It is one of the largest publicly available **labeled** cryptocurrency transaction datasets, making it a recognized benchmark in AML/fraud-detection research
- It contains genuine real-world data challenges (severe imbalance, ~77% unlabeled nodes, temporal dependency) that justify meaningful preprocessing and modeling decisions rather than working with an already-clean toy dataset

**Scale:** 203,769 transactions (nodes), 234,355 directed edges (payment flows), spanning 49 time steps (~2-week intervals). ~21% labeled licit, ~2% labeled illicit, ~77% unlabeled.

## 3. Literature Review

1. Weber, M., Domeniconi, G., Chen, J., Weidele, D.K.I., Bellei, C., Robinson, T., & Leiserson, C.E. (2019). *Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics.* arXiv:1908.02591. — Introduces the Elliptic Data Set itself and demonstrates that graph-based features improve illicit-transaction detection over purely local features.
2. Alarab, I., Prakoonwit, S., & Nacer, M.I. (2020). *Competence of Graph Convolutional Networks for Anti-Money Laundering in Bitcoin Blockchain.* — Explores GCN-based approaches applied to the same Elliptic dataset, useful for comparing graph-based vs. tabular approaches.
3. (GuiltyWalker paper, arXiv:2102.05373) — Proposes a distance-to-illicit-nodes feature engineering approach on the Elliptic graph, relevant for justifying feature engineering choices beyond the raw 166 features.
4. (Elliptic++ Dataset paper, arXiv:2306.06108) — Extends the original Elliptic dataset with wallet-level actor data; useful for discussing dataset limitations and future scope in your DA3 conclusion.
5. (Detecting Anomalous Cryptocurrency Transactions paper, arXiv:2206.04803) — Applies ML-based forensics techniques to the same dataset; useful for comparing algorithm choices and evaluation metrics.

*(Search each title on Google Scholar or arXiv to pull the full citation format your department requires — APA/IEEE — before submitting.)*

## 4. Dataset Understanding and Feature Description

The dataset consists of three files:

- **elliptic_txs_features.csv** — 203,769 rows × 167 columns: transaction ID, time step, and 165 anonymized features (94 local transaction-level features such as number of inputs/outputs and transaction fee proxies; 71 aggregated features summarizing the 1-hop neighborhood of each transaction — max, min, std, correlation coefficients)
- **elliptic_txs_classes.csv** — label per transaction: `1` (illicit), `2` (licit), `unknown`
- **elliptic_txs_edgelist.csv** — directed edges representing Bitcoin flow between transactions, defining the graph structure

All features are anonymized (Elliptic does not disclose exact real-world meaning for confidentiality reasons) — this is itself worth noting as a limitation in your report.

## 5. Data Preprocessing

1. **Merge** features + classes on `txId`
2. **Filter** out `unknown`-labeled rows, keeping only the ~46,564 labeled transactions for supervised learning
3. **Time-aware split**: train on time steps 1–34, test on time steps 35–49 (avoids leaking future information into training)
4. **Class balancing**: apply SMOTE (Synthetic Minority Over-sampling Technique) to the *training set only*, bringing the illicit class up to a more workable ratio without touching the test set
5. **No missing-value imputation needed** — Elliptic's features are pre-computed and complete, but note this explicitly in your report rather than skipping the step silently

## 6. Exploratory Data Analysis — 5 Visualizations (R / ggplot2)

1. Class distribution before vs. after SMOTE (bar chart)
2. Feature density comparison — licit vs. illicit (density plot)
3. Temporal transaction volume across 49 time steps, split by label (line chart)
4. Correlation heatmap of the top 15 features most associated with the illicit label
5. Local feature vs. aggregated neighborhood feature scatter plot with trend line

*(See accompanying R script: `DA1_elliptic_fraud_pipeline.R` — generates all 5 plots automatically once you point it at your downloaded CSVs.)*

## 7. Tools

- **DA1:** R (`readr`, `dplyr`, `ggplot2`, `themis` for SMOTE)
- **DA2:** PostgreSQL for structured storage, Python (`scikit-learn`, `xgboost`, `lightgbm`) for model training
- **DA3:** Streamlit dashboard with SHAP-based feature explanations